ARCHITECTURAL BLUEPRINT & IMPLEMENTATION GUIDE

DevDeck v2 — Full Technical Specification

Complete component architecture, interaction behaviors, and implementation roadmap prepared for Claude Opus engineering execution.

PROJECT	TARGET MODEL	ARCHITECTURE	STATUS
DevDeck v2	Claude Opus 4.6	Chrome Ext MV3 / Vite	Approved & Frozen

1. Executive Summary & Philosophy

DevDeck v2 elevates the developer browser startup cockpit into a state-of-the-art dashboard blending rich developer telemetry (real-time GitHub contribution matrix, local weather & UV index, cron task reminders) with keyboard-first bookmarks, smart autocomplete search, and live tech intelligence.

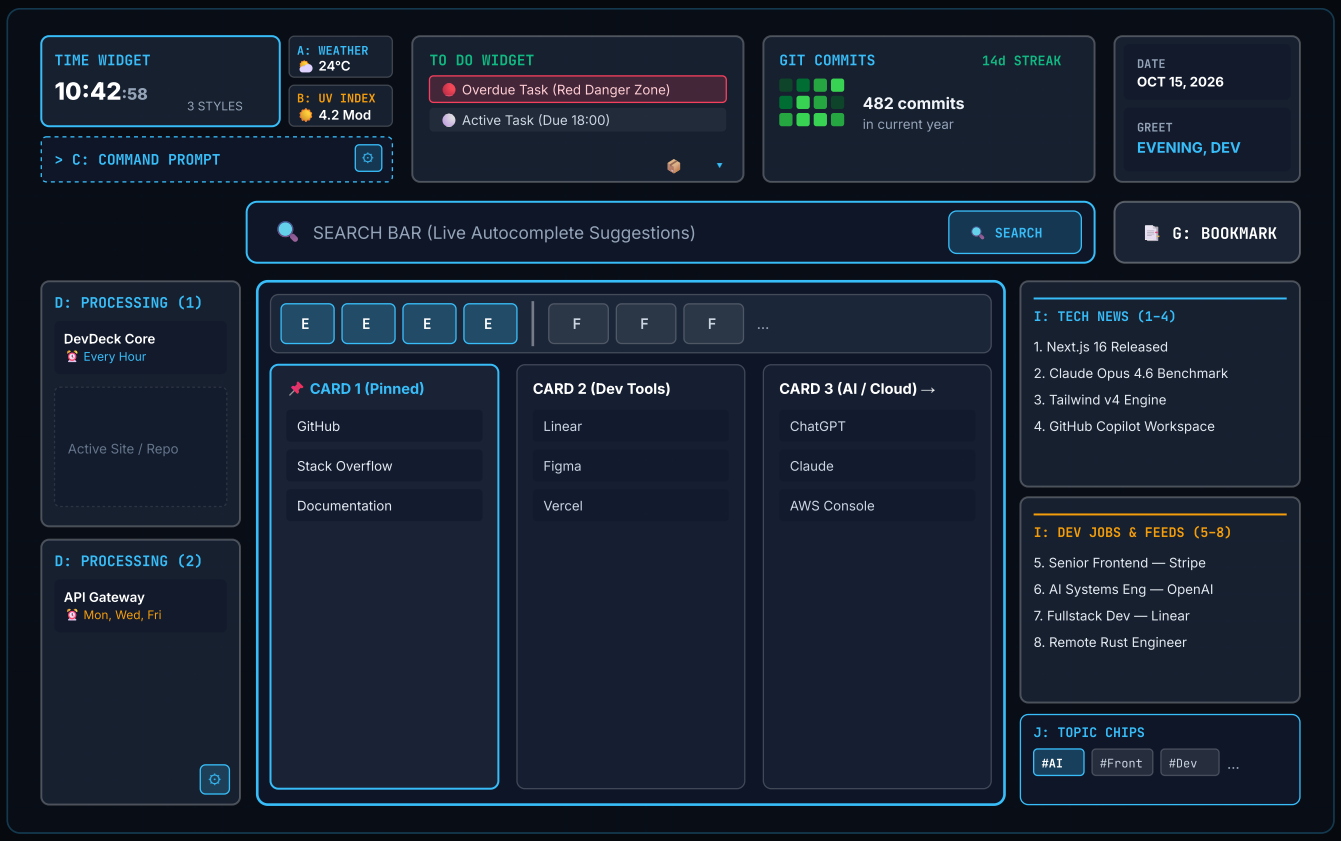
The system adheres to an **Apple-grade minimalist aesthetic** with Obsidian Glass styling, `backdrop-filter: blur(24px)`, custom cubic-bezier transitions, and zero mandatory backend dependencies (pure client-side Manifest V3 Chrome Extension compatibility).

2. Dashboard Layout & Spatial Architecture

The dashboard grid strictly maps to the user's hand-drawn schematic (`media_1792007261268.jpg`). The lower half of the screen is structured as a **3-column cockpit** where **Quick Links (E) & Recent Links (F)** are integrated *inside the top header of the Center Card Container*, flanked by **Currently Processing (D)** on the left and **Scraped Intelligence (I) & Topics (J)** on the right:

FIG 1.0 — DEVDECK V2 ARCHITECTURAL BLUEPRINT (1:1 SKETCH RECONSTRUCTION)

1920 × 1080 VIEWPORT



Clock TIME
DSEG7 / Cyber / macOS

Weather A

UV B

C: > Terminal Prompt ⚙ Admin

To Do Task Manager TO DO
Overdue red danger zone, 30s auto-shrink overlay, bottom-right archive icon 📦 & expand ▾

Git Commits Heatmap GIT
Classic green contribution tiles, current & longest streak, annual commit tally

DATE/GREET
WED, OCT 15
GOOD EVENING

Always-Visible Search Bar (Live Google Autocomplete) SEARCH
🔍 SEARCH

Bookmarks / Site Map G

⚙ Processing (1) D
Active target site with cron schedule

⚙ Processing (2) D
Active target site with cron schedule

E E E **E (Fixed 1-4)** | F F **F (Recent Links → scroll)**

📌 CARD 1 (Pinned)
Anchored to 1st place while pinned

CARD 2
Scrolls horizontally

CARD 3 →
Scrolls horizontally

News (1–4) I
Top breaking tech articles

Jobs (5–8) I
Dev roles & hiring feeds


Mini Settings
Topics
J
#AI
#Front
#Dev

3. Component-by-Component Detailed Specifications

3.1. TIME Widget (Top-Left)

Provides primary temporal telemetry with 3 distinct user-selectable styles in Settings → Appearance:

STYLE VARIANT	VISUAL CHARACTERISTICS	USER CONSTRAINTS & SPECS
1. Existing Digital (DSEG7)	Classic 7-segment digital font, glowing active segments with faint unlit backdrop segments, pulsing colon.	Preserve existing 12/24h toggle and accent color glow.
2. Cyberpunk 2077	Massive, ultra-bold, condensed futuristic display font. Monospace `// WORLD TIME` header top-left, matrix dots top-right.	**STRICT RULE:** Do NOT show multiple timezone cities below (LA, New York, London, Paris, Kyiv). Render only massive Hours/Minutes with small, half-height Seconds on the right flank.
3. macOS Plain Digital	Clean San Francisco / Inter Apple-style typography (weights 300/400). Minimalist glass backing, tabular numbers.	Subtle specular highlights, elegant modern aesthetic.

3.2. A & B: Local Weather & UV Index Widgets

Positioned immediately adjacent to the TIME widget in a split glass card:

- **Component A (Weather):** Displays temperature in Celsius (°C) and weather condition icon (Sun, Cloud, Rain, Snow). Uses the browser's HTML5 Geolocation API with graceful fallback to manual coordinates/city. Fetches from the open Open-Meteo Forecast API (no API key required).
- **Component B (UV Index):** Displays the solar ultraviolet radiation index with color-coded risk badge: `0-2 Low (Green)`, `3-5 Mod (Yellow)`, `6-7 High (Orange)`, `8-10 Very High (Red)`, `11+ Extreme (Purple)`.

> 📡 **Network Strategy**
> Data is cached in `localStorage` for 45 minutes to prevent redundant network queries on every new tab open.

3.3. C: Command Prompt & Terminal Launcher

A slim, responsive CLI input positioned directly below the TIME and Weather/UV widget box.

- **Input Field:** Monospace prompt (`> enter command or shortcut...`) with tab completion.
- **System Administrator Icon:** A small wrench-and-gear / admin shield icon (matching reference image `media_1792009911738.png`) located on the right side of the input.
- **Elevated Admin Execution:** When the admin icon is toggled active (glowing amber/red ring), pressing Enter triggers the terminal launch in elevated / Administrator mode (PowerShell on Windows, Bash/Sudo on Linux).
- **In-Browser Emulated Commands:** Built-in instant shortcuts: `gh <repo>` , `npm <pkg>` , `open <url>` , `calc <math>` , `b64e/b64d` .

3.4. TO DO Widget with Overdue Red Danger Zone & Auto-Shrink

Centrally positioned top-row productivity manager featuring four critical interaction behaviors:

> 🚨 **Red Danger Zone Escalation**
> If a to-do item exceeds its specified Due Date and Due Time (`now > task.dueTimestamp`), it is automatically pinned to the very top of the list and highlighted with an animated crimson danger background (`rgba(239, 68, 68, 0.25)`).

> ⏱️ **30-Second Inactivity Auto-Shrink**
> Clicking the expand button in the bottom-right corner reveals the full task list as a high-z-index overlay over the content below. **If the user does not interact with the list for 30 seconds, it automatically shrinks back to its collapsed preview state.** Any click or keystroke resets the 30-second timer.

- **Bullet Completion to Archive:** Clicking the leading circular bullet marks the task complete with a slide animation, moving it into the Archive.
- **Archive Drawer Icon:** A mini settings/archive icon positioned **immediately to the left of the expand button** in the bottom-right corner allows viewing, restoring, or purging archived tasks.

3.5. GIT COMMITS Widget (GitHub Activity Matrix)

Displays a live GitHub contribution graph styled with classic green tiles:

- **Contribution Heatmap:** 16-to-52 week compact grid with 5 level shades of emerald green.
- **Telemetry Metrics:** Current Active Streak (days), Longest Streak (days), and Total Commits in current calendar year.
- **Settings Integration:** Accepts user's GitHub username and optional Personal Access Token (PAT) for private repo visibility.

3.6. Search Bar & G_1 Visual Site Map Browser

- **Always-Visible Search Bar:** Centered horizontal anchor. Includes an integrated **[🔍 SEARCH]** button inside the right edge of the input.
- **Live Autocomplete Suggestions:** As the user types, a multi-tier dropdown provides matching saved bookmarks, Google web autocomplete predictions, and direct search engine query triggers.
- **G_1 Bookmark Button & Site Map:** Located outside to the right of the search bar. Clicking opens an elongated panel with a visual tree site map enabling users to:
  - Drag-and-drop or move sites between category folders.
  - Add, rename, or delete category folders.
  - Instant live search across all nested bookmarks.

3.7. E (Quick Links) & F (Recent Opened Links) Strips

- **Pinned First 4 Links FIXED:** The first 4 links in Strip E are permanently anchored on the left and do not scroll.
- **Auto-Populated Links:** Positions 5 onward are dynamically computed based on visit frequency (`visits`) and recency (`lastVisited`). The count is adjustable via a slider in Settings (2 to 12).
- **Horizontal Scroll:** Pinned links stay fixed while auto-picked links scroll horizontally underneath a gradient fade mask.
- **Recent Strip (F):** Separated from Quick Links by a clean glass divider (`|`). Displays recently opened destinations as chips with favicons and relative visit timestamps.

3.8. D: Currently Processing (Left Sidebar & Cron Engine)

A dedicated vertical panel for active workspaces and repositories under active development:

- Holds N active project sites.
- **Mini Settings Cog:** Dedicated button at the bottom of the panel to add/remove active sites.
- **Per-Site Cron Reminder Schedule:**

SCHEDULING PARAMETER	SUPPORTED OPTIONS
Frequency	Hourly, Daily, or Custom Interval
Weekly Filter	Specific Days of the Week (Mon, Tue, Wed, Thu, Fri, Sat, Sun)
Active Date Span	From Month-Year to Month-Year (e.g. 2026-10 to 2026-12)
Notification Delivery	Browser Notification / `chrome.alarms` API with 1-click tab launch

3.9. Main Category Cards — Single-Card Pinning Mechanism

Center stage horizontal scrollable card stack with strict pinning rules:

> 📌 **Single-Card Pin Constraint**
> **User can pin EXACTLY ONE category card at any time.** Pinning a card moves it temporarily to the FIRST position in the row and anchors it during horizontal scroll. Unpinning immediately restores the card to its original sequenced position.

3.10. I & J: Tech News, Job Intelligence & Topic Chips (Right Sidebar)

- **Dual-Section News Feed (I):** Upper section renders items 1–4 (breaking news/repos); lower section renders items 5–8 (developer jobs/trending). Sources include HackerNews, Dev.to, GitHub Trending, and RemoteOK.
- **Numbered List & Sliders:** Each panel has a top accent slider bar and items numbered 1–4 and 5–8. Clickable destination links with source badge and relative timestamp.
- **Topic Filter Chips (J):** Interactive pills (`#AI` , `#Frontend` , `#DevOps` , `#Security` , `#Career`) at the bottom of the sidebar. Clicking any chip immediately filters the news feed.
- **Dedicated Scraping Settings:** Full CRUD in Settings to manage source RSS feeds and topic keywords.

3.11. ⚙️ Central Settings Modal Architecture

The main Settings overlay is expanded into a multi-tab developer preferences center:

SETTINGS TAB	CONFIGURATION CONTROLS & FIELDS
1. Appearance	Clock Style Picker (DSEG7, Cyberpunk, macOS), 12h/24h toggle, Show seconds toggle, Accent colors (Cyan, Emerald, Amber, Violet, Rose), Background wallpaper (Image URL, presets, blur, opacity).
2. Profile & Git	User greeting name (for top-right greeting), GitHub username (for contribution tiles matrix), optional GitHub Personal Access Token (PAT).
3. Weather & UV	Auto-detect location toggle (GPS / Geolocation), Manual city / coordinate override, Celsius/Fahrenheit toggle, Cache refresh duration.
4. Currently Processing	Manager for active project workspaces. Per-site cron reminder schedule: Frequency (hourly, daily, custom), Weekday filters (Mon-Sun), Active date span (Month-Year to Month-Year).
5. Quick & Recent Links	Selection of the 4 permanently fixed Quick Links, Auto-picked links count slider (2 to 12), Clear recent history action.
6. Bookmarks & Site Map	Visual Site Map manager, category folder hierarchy editor, drag-and-drop site moving, JSON data backup & restore.
7. Scraping & News	Manage custom RSS/news endpoints, configure topic filter tags (CRUD for `#AI`, `#DevOps`, etc.), toggle dev jobs feed visibility.

4. Complete TypeScript Domain Schemas for Claude Opus

```
// Core Domain Models for DevDeck v2
export type ClockStyle = 'dseg7' | 'cyberpunk' | 'macos';

export interface WeatherData {
  temperature: number; // in Celsius
  conditionCode: number;
  conditionText: string;
  city: string;
  uvIndex: number;
  uvRisk: 'low' | 'moderate' | 'high' | 'very-high' | 'extreme';
  lastUpdated: number;
}

export interface TodoItem {
  id: string;
  text: string;
  dueDate?: string; // YYYY-MM-DD
  dueTime?: string; // HH:mm
  dueTimestamp?: number; // unix epoch ms
  isOverdue: boolean; // true if now > dueTimestamp
  completed: boolean;
  completedAt?: number;
  archived: boolean;
  priority: 'low' | 'normal' | 'urgent';
}

export interface GitStats {
  username: string;
  totalCommitsYear: number;
  currentStreak: number;
  longestStreak: number;
  contributions: {
    date: string;
    count: number;
    level: 0 | 1 | 2 | 3 | 4;
  }[];
  lastFetched: number;
}

export interface ProcessingSite {
  id: string;
  name: string;
  url: string;
  icon?: string;
  notes?: string;
  schedule: {
    enabled: boolean;
    frequency: 'hourly' | 'daily' | 'weekly';
    daysOfWeek: number[]; // 0=Sun, 1=Mon, ..., 6=Sat
    timeOfDay?: string; // HH:mm
    startMonthYear: string; // YYYY-MM
    endMonthYear: string; // YYYY-MM
  };
}

export interface Category {
  id: string;
  name: string;
  icon: string;
  order: number;
  isPinned?: boolean; // EXACTLY ONE category card may be pinned at a time
}
```

```ts
export interface SiteBookmark {
  id: string;
  title: string;
  url: string;
  icon?: string;
  categoryId: string;
  visits: number;
  lastVisited: number;
  isQuickLinkPinned?: boolean; // for first 4 fixed links
}

export interface NewsItem {
  id: string;
  title: string;
  url: string;
  source: 'HackerNews' | 'Dev.to' | 'GitHub' | 'RemoteOK';
  category: 'tech' | 'job' | 'trending';
  tags: string[];
  publishedAt: number;
}

export interface TopicChip {
  id: string;
  label: string; // e.g. "AI"
  tagQuery: string; // e.g. "ai,llm"
}

export interface AppSettings {
  clockStyle: ClockStyle;
  timeFormat24h: boolean;
  showSeconds: boolean;
  accentColor: string;
  backgroundType: 'gradient' | 'image';
  backgroundImageUrl: string;
  backdropBlurPx: number;
  userName: string;
  githubUsername: string;
  githubToken?: string;
  weatherEnabled: boolean;
  useGeolocation: boolean;
  manualCityName?: string;
  searchEngine: 'google' | 'duckduckgo' | 'github' | 'perplexity';
  quickLinksAutoPickCount: number; // 2 to 12
  enabledTopics: string[];
}
```

5. Claude Opus Implementation Roadmap

STAGE	PRIMARY DELIVERABLES	KEY TARGET FILES
Stage 1	Data models & Zustand store slices (Todos, Weather, Cron, Single-Pin)	`src/types.ts`, `src/store/useStore.ts`
Stage 2	Top Telemetry Bar: 3 Clock styles, Weather & UV, Admin Terminal Prompt, Overdue Todo with 30s auto-shrink, Git Heatmap	`src/components/Clock/*`, `WeatherWidget.tsx`, `TodoWidget.tsx`, `GitWidget.tsx`
Stage 3	Search Bar with inner button, Google autocomplete, Gᵢ Bookmark Site Map	`src/components/Search/SearchBar.tsx`, `SiteMapModal.tsx`
Stage 4	Quick Links (4 fixed + scrollable auto-picked) & Currently Processing left sidebar with cron reminders	`QuickLinksStrip.tsx`, `CurrentlyProcessingPanel.tsx`
Stage 5	Category Cards single-pin state, News dual panel with Topic chips, Settings modal expansion	`CategoryCard.tsx`, `ScrapedNewsPanel.tsx`, `SettingsModal.tsx`

✅ **Specification Verified**

This document constitutes the frozen, complete specification for DevDeck v2. All architectural questions and user constraints are fully resolved.